

(a) We have

$$2 + 5 + 8 + \dots + 3n - 1 = \frac{n(3n+1)}{2}$$

Rewritten using summation gives us

$$\sum_{i=1}^n 3i - 1 = \frac{n(3n+1)}{2}$$

Proof

BASE CASE:

Let $n = 1$

$$\begin{aligned}\sum_{i=1}^n 3i - 1 &= \sum_{i=1}^1 3(1) - 1 = 3 - 1 \\ &= \boxed{2}\end{aligned}$$

$$\begin{aligned}\frac{n(3n+1)}{2} &= \frac{1 \cdot (3(1) + 1)}{2} = \frac{3 + 1}{2} \\ &= \frac{4}{2} \\ &= \boxed{2}\end{aligned}$$

$$\begin{aligned}2 &= 2 \\ \therefore n = 1 \text{ is true}\end{aligned}$$

INDUCTIVE CASE:

Assume that

$$\sum_{i=1}^n 3i - 1 = \frac{n(3n+1)}{2}$$

We must prove

$$\begin{aligned}\sum_{i=1}^{n+1} 3i - 1 &= \frac{(n+1)(3(n+1) + 1)}{2} \\ &= \frac{(n+1)(3n+4)}{2}\end{aligned}$$

$$\begin{aligned}\sum_{i=1}^{n+1} 3i - 1 &= \sum_{i=1}^n 3i - 1 + (3(n+1) - 1) \\&= \frac{n(3n+1)}{2} + 3(n+1) - 1 \\&= \frac{n(3n+1)}{2} + 3n + 2 \\&= \frac{n(3n+1)}{2} + \frac{6n+4}{2} \\&= \frac{n(3n+1) + (6n+4)}{2} \\&= \frac{3n^2 + n + 6n + 4}{2} \\&= \frac{3n^2 + 7n + 4}{2} \\&= \frac{(n + \frac{3}{3})(n + \frac{4}{3})}{2} \\&= \boxed{\frac{(n+1)(3n+4)}{2}}\end{aligned}$$

$$\frac{(n+1)(3n+4)}{2} = \frac{(n+1)(3n+4)}{2} \checkmark$$

(b) We have

$$\prod_{k=1}^n \left(1 + \frac{1}{k}\right) = n + 1$$